Chatbot Project Workflow & Tools Documentation

1 Workflow Overview

Step 1: Dataset Preparation

- **Tool:** JSON
- **Action:** Create a dataset of intents, patterns, and responses (e.g., `clothes_data.json`).
- **Purpose:** The chatbot needs labeled data to learn which response matches a user input.

Step 2: Text Preprocessing

- **Tool:** Keras (`Tokenizer`, `pad_sequences`)
- **Action:**
    - Convert sentences to **lowercase**
    - Tokenize words → convert text → numbers
    - Pad sequences → make all sequences same length for LSTM input
- **Purpose:** Neural networks only understand numbers, not text.

Step 3: Build Model

- **Tool:** TensorFlow + Keras
- **Action:**
    - **Sequential model** with layers:
        - **Embedding** → convert words → vectors

        - **LSTM** → learn sequences in sentences
        - **Dropout** → prevent overfitting
        - **Dense + Softmax** → predict intent probabilities
- **Purpose:** Train a model to classify user messages into correct intents.

Step 4: Train & Save Model

- **Tool:** Keras (TensorFlow backend)
- **Action:**
    - Train model on sequences and labels
    - Save the model in **Keras 3 format** (`shop_bot.keras`)
    - Save **tokenizer** and **classes** using `pickle`
- **Purpose:** Keep a trained model ready for prediction without retraining.

Step 5: Load Model for Inference

- **Tool:** TensorFlow + Pickle
- **Action:**
    - Load model (`shop_bot.keras`)
    - Load tokenizer and classes
    - Preprocess user input → predict intent → get response
- **Purpose:** Use trained model in production/chat interface.

Step 6: Build Chat Interface

- **Tool:** Gradio (HuggingFace)
- **Action:**
    - `gr.ChatInterface()` → provide chatbot function
    - Launch app locally or with `share=True` for public link
- **Purpose:** Provides **interactive web UI** for users to chat with your bot.

Step 7: Deploy on Website

- **Tool:** JavaScript (with HTML/CSS)
- **Action:**
  - Create floating chat window
  - Use **Gradio's hosted link or API** to send/receive messages
  - Update chatbox dynamically with user and bot messages
- **Purpose:** Integrate chatbot into **live website**.

2️⃣ Tools & Their Roles

Tool / Library	Role in Project	Example Usage
TensorFlow/Keras	Build, train, save neural network	`Sequential`, `LSTM`, `Dense`
JSON	Store chatbot dataset	`clothes_data.json`
Pickle	Save/load Python objects	`pickle.dump(tokenizer)`
Gradio (HuggingFace)	Web-based chat UI	`gr.ChatInterface(fn=chatbot)`
JavaScript	Integrate bot into website	Floating chat window, send API requests

3️⃣ Summary

1. Prepare dataset → JSON
2. Preprocess text → Keras Tokenizer + pad_sequences
3. Build model → Sequential (Embedding → LSTM → Dropout → Dense)
4. Train & save model → TensorFlow/Keras
5. Load model for inference → predict intent → select response
6. Chat interface → Gradio
7. Deploy to website → JavaScript connects to Gradio link

✅ **Workflow Diagram (Simplified)**

```
User Input
    ↓
JavaScript Chat Window
    ↓
Gradio Hosted Bot / API
    ↓
Preprocess Text → Tokenizer + pad_sequences
    ↓
Sequential Model (Embedding → LSTM → Dropout → Dense)
    ↓
Softmax → Predicted Intent
    ↓
Select Response from JSON
    ↓
Send Response Back to JavaScript → Display to User
```

This covers a **full end-to-end flow**, tools, and their usage in your chatbot project.

If you want, I can also **create a small visual diagram version** of this workflow that you can use in documentation or presentations.

Do you want me to do that?